

Unit

2

Lesson

1



Innovators in Action.

Feel the Force.

Time Frame: [90 min]

Lesson Objectives:

By the end of this lesson students will be able to:

- Explain Newton's Third Law in their own words and apply it to real-world movement (e.g., walking with a walker).
- Identify action-reaction force pairs in the context of human motion and device use.
- Analyze balance and movement using force diagrams.
- Collaborate with peers to explore and explain physical forces in everyday design.

Challenge questions of the lesson

- If you push harder on a walker, does it move faster, and what happens to your body in return? Why?
- What might happen if the walker doesn't push back with equal force (e.g., weak wheels or slippery grips)?
- How do action and reaction forces help keep an elderly person balanced when using a walker?
- How could understanding these forces help improve the design of safer mobility aids?
- If the ground were slippery or uneven, how would that affect balance? What real-world examples can you think of?

SEL Relation

- **Self-Awareness:** Students begin to recognize their own balance and movement, comparing it to how others (like elderly individuals) experience walking differently. They become more conscious of physical limitations and their emotional effects.
- **Social Awareness:** By thinking from the perspective of someone using a walker, students build empathy. They consider what it means to struggle with basic movement—and how design can restore dignity, confidence, and safety.
- **Responsible Decision-Making:** Students connect scientific understanding with the ethical responsibility of designing for real people. They learn to value inclusive design and think critically about how science can solve human problems.
- **Relationship Skills:** In discussion or paired activities, students practice active listening, collaboration, and respect for differing perspectives—key in both design and teamwork.

Engage



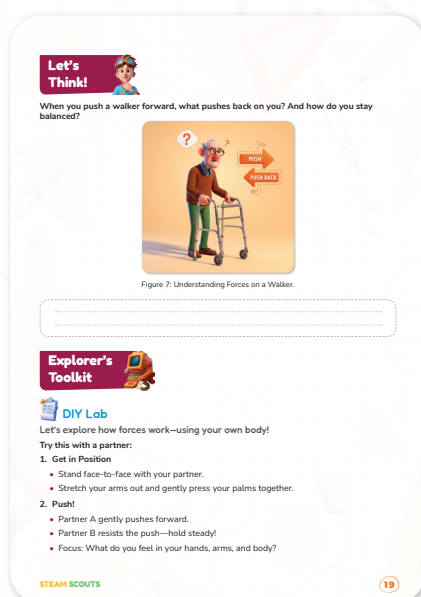
Teacher's Role:

- **Facilitates Student Exploration:** Guides students' initial thoughts by asking open-ended questions and encouraging them to share their prior experiences.
- **Explains Core Scientific Principles:** Teaches fundamental concepts in a clear and accessible way, specifically:
 - **Newton's Third Law:** The principle of action and reaction, where pushing the walker results in an equal and opposite force back on the person.
 - **Force of Friction:** Explains that the forward motion comes from the friction between the person's shoes and the floor.
- **Connects Concepts to Context:** Links these scientific ideas to the students' daily lives to make the learning relevant and memorable.

Expected From Students:

- **Engage in Initial Hypothesizing:** Students are expected to think critically about the question and propose their own initial ideas or hypotheses.
- **Participate in Discussion:** Actively listen to their peers and the teacher, contributing their own thoughts and questions.
- **Grasp and Apply New Concepts:** Demonstrate an understanding of the concepts of Newton's Third Law, friction, and center of gravity by discussing them.
- **Formulate a Written Response:** Synthesize their learning into a clear and concise written answer that addresses both parts of the original question, showing their comprehension.

Parts of the book included.



Let's Think!



Objective:

The student will be able to explain the forces involved when pushing an object, specifically applying Newton's Third Law and the concept of friction. They will also be able to describe how a walker's design provides stability by widening the base of support and keeping a person's center of gravity balanced. This will demonstrate their ability to connect scientific principles to the design of an everyday object.

Possible Answers:

Science-Based Answers (Newton's Third Law):

- The walker pushes back on you with equal force in the opposite direction.
- Your feet push backward on the ground, and the ground pushes you forward to keep you balanced.
- The ground provides a reaction force that supports your body so you don't fall.
- Your body adjusts its center of gravity to stay upright while moving the walker.
- Friction between your feet and the floor helps you avoid slipping when the walker moves forward.

Student-Friendly, Realistic Responses:

- "The walker kind of pushes back on me—it doesn't just slide away."
- "I feel like I'm leaning forward, so I have to hold myself up."
- "My legs push into the ground to keep me steady."
- "If the walker moves too fast, I could fall, so I hold back a bit."
- "I have to shift my weight to stay standing when I push it."

Misconceptions You Might Hear:

- "Nothing pushes back." → Gently guide them to observe the reaction force from the walker.
- "Only the person is doing something." → Prompt them to think about how movement involves interaction between objects.

Possible Strategies for Implementation:

Strategy 1: Think-Pair-Share

Implementation:

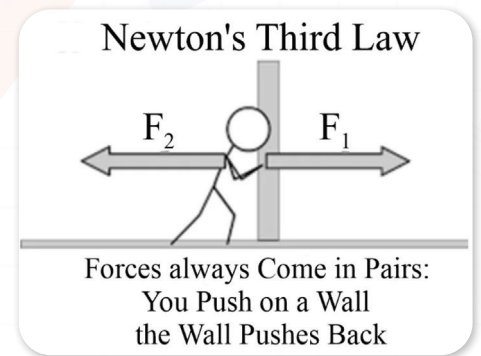
- **Step 1 (Think):** Ask students to silently reflect for 1–2 minutes and jot down their thoughts in a notebook or sticky note.
- **Step 2 (Pair):** Pair them up to discuss what they each wrote. Encourage them to challenge or build on each other's ideas.
- **Step 3 (Share):** Have a few pairs share their ideas with the whole class. Write their key points on the board to create a class idea bank.

Goal: Activate prior knowledge and get all students engaged even those who are more reflective or shy.

Strategy 2: Use a Quick Demonstration or Model

Implementation:

- **Push a classroom chair or cart forward and ask:** “What just happened? What pushed back on me?”
- **Draw two arrows:** one from your hands (pushing forward), one from the chair (pushing back on you).
- **Link this to Newton’s Third Law:** “Every action has an equal and opposite reaction.”



Goal: Make the invisible concept (force) visible and relatable.

Strategy 3: Visual Thinking Prompt

Implementation:

- Show or sketch a simple force diagram on the board:
 - Person → pushing walker (action force)
 - Walker → pushing back on person (reaction force)
 - Ground → pushing up on feet (balance support)
- Ask: “What would happen if one of these forces wasn’t there?”

Goal: Use visual support for learners who benefit from diagrams and structured thinking.

Strategy 4: Journaling with an Empathy Angle

Implementation:

- **Add a SEL twist:** “Imagine you're your grandpa. What does it feel like to rely on a walker to stay balanced?”
- Students write a short paragraph from the perspective of an elderly user experiencing these forces daily.

Goal: Combine critical thinking with empathy-building to prepare for user-centered design.

Explorer's Toolkit



Teacher's Role:

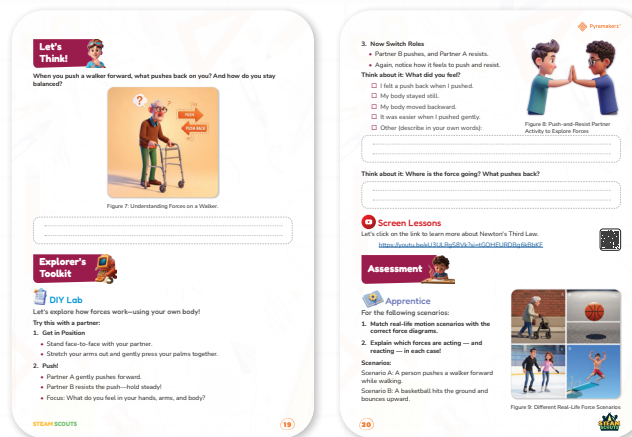
- **Sets the Stage:** The teacher introduces the activity and ensures students understand the instructions and the goal of exploring forces.
- **Keeps Everyone Safe:** The teacher reminds students to push gently to avoid accidents.
- **Guides the Learning:** The teacher walks around, watches the students, and asks simple questions to help them think about what they are feeling.
- **Leads the Discussion:** After the activity, the teacher brings the class together to talk

about their observations and helps them connect their experience to the scientific idea of forces pushing back.

Expected From Students:

- **Follow the Steps:** Students work with a partner to complete the pushing activity exactly as it is written on the worksheet.
- **Pay Attention to Feelings:** They need to pay close attention to what they feel in their hands and body during the push.
- **Think and Talk:** Students think about the questions and are ready to share what they felt with their partner and the rest of the class.
- **Write Their Answers:** They check the correct boxes and write down their thoughts on the worksheet.

Parts of the book included.



DIY Lab

Objective:

Students will be able to demonstrate an understanding of action-reaction forces by participating in a hands-on activity with a partner. They will learn to identify and describe the feeling of balanced forces, applying the concept of Newton's Third Law to explain that when they push, their partner pushes back with an equal force. Ultimately, they will be able to connect this physical experience to the fundamental scientific principle of forces.

Materials needed:

- A Partner: This activity requires students to work in pairs.

Activity Instructions:

- **Pair Up:** Students find a partner and stand facing each other.
- **Gentle Push:** One partner (Partner A) gently pushes on the other's palms while the second partner (Partner B) resists the push.
- **Focus on Sensation:** Students focus on the feelings in their hands, arms, and body during this interaction. They should feel the force pushing back.
- **Switch Roles:** The partners switch roles, with Partner B pushing and Partner A resisting.

- **Reflect and Record:** After the activity, students individually complete the "Think about it" sections on the worksheet by checking the appropriate boxes and writing their explanations.

Possible Strategies for Implementation:

Part 1: Push and Resist Challenge (Action-Reaction Forces)

Strategy 1: Kinesthetic Pair Activity

Implementation:

- Students pair up and stand facing each other.
- Hands touching, they gently push.
- One partner push slightly; the other resists.
- They switch roles after a few tries.

Teacher Language (integrating SEL + UDL):

- "Feel what's happening — not just with your hands, but your whole body!"
- Your safe partners here, so communicate: 'Pushing now!' or 'Too much!'
- Let's make sure everyone feels comfortable and respected."

Strategy 2: Visual and Verbal Reflection

Implementation:

- Pause and ask:
 - "What did you notice when you pushed?"
 - "Did your body stay still or move? Why?"
- Students choose ONE way to explain... they can say their observation, draw it, or act it out (pantomime the forces).

Varieties Offered (UDL):

- Quick drawing → Draw a picture of you and your partner pushing, with arrows showing forces.
- verbal explanation. → Write 2-3 sentences about what happened.
- Physical mimic (mime pushing + resisting). → ☐ Act it out in a group and explain in words.

Teacher Prompt (SEL built-in):

- "Everybody might notice forces a little differently — that's the beauty of science! Let's listen carefully to each other's discoveries."

Reflection Questions:

- How did it feel working with a partner today?
- What surprised you most about forces?
- Write or draw your answer below:

Explain



Teacher's Role:

- **Curator and Facilitator:** The teacher selects the video as a core resource to introduce Newton's Third Law. Their role is to facilitate understanding by leading discussions and activities that build upon the video's content.
- **Context Provider:** The teacher provides the "why" behind watching the video, explaining that it will introduce a key physics concept.
- **Discussion Leader:** The teacher will lead the class in a structured discussion before, during, and after the video to ensure comprehension.
- **Concept Reinforcer:** After the video, the teacher connects the video's examples (swimmer, rocket) to other real-world scenarios or a hands-on activity to solidify the students' understanding.

Expected from Students:

- **Active Viewing:** Students are expected to watch the video with a specific purpose in mind, rather than passively. This involves listening for key definitions and examples.
- **Note-Taking:** Students should take notes on the definition of Newton's Third Law and a few of the examples provided in the video.
- **Critical Thinking and Questioning:** Students are expected to think about the content and be prepared to ask questions or offer their own examples during class discussions.
- **Participating in Discussion:** Students will contribute to the pre-watching, during-watching, and post-watching discussions by sharing their thoughts and answering questions.

Parts of the book included.



Screen Lessons

Let's click on the link to learn more about Newton's Third Law.

<https://youtu.be/eU3ULRgS8Vk?si=tGOHEURDBg6kBbKF>



Screen Lessons

Objective:

Students will be able to define Newton's Third Law of Motion and provide several real-world examples of action-reaction force pairs from the video. They will also be able to explain how these forces, in conjunction with an object's mass, determine its acceleration.

Video Link: <https://youtu.be/eU3ULRgS8Vk?si=tGOHEURDBg6kBbKF>



Big Ideas:

- $\text{Work} = \text{Force} \times \text{Distance}$
- $\text{Power} = \text{Work} \div \text{Time}$
- Efficiency = Doing the same work with less effort or in less time

Possible Strategies for Implementation:

1. Pre-Watching: Set a Purpose

- **Hook Question:** Start with a question to pique student interest, such as "What happens when you push off a wall in a swimming pool?" or "Why does a rocket launch upward?"
- **Define the Goal:** Tell students they are going to watch a video to learn the science behind these questions. Instruct them to pay attention to the definition of Newton's Third Law and to note at least two examples.

2. During Watching: Active Engagement

- **Pause and Discuss:** The teacher can pause the video at key points (e.g., after the swimmer example, or after the rocket example) to check for understanding and have students briefly discuss the force pairs they just saw.
- **Guided Notes:** Provide students with a simple worksheet or template to fill out as they watch, with spaces for the definition and each example.

3. Post-Watching: Discussion and Reflection

- **Summary & Review:** Begin with a class discussion where students share the definition and examples they noted.
- **Application Activity:** Challenge students to come up with their own unique examples of Newton's Third Law that were not in the video. This could be a quick whiteboard activity or a small-group brainstorm.
- **Connect to Prior Knowledge:** Link this video to the "DIY Lab" activity from the previous turn, reinforcing that their hands-on experience of pushing a partner was a perfect example of this law.

Elaborate and Evaluate



Teacher's Role:

- **Lab Supervisor:** For the "Builder" activity, the teacher acts as a supervisor, ensuring that students are using the spring scales and timers correctly and safely. They will circulate the room to provide guidance on the experimental procedure and data collection.
- **Data Analysis Coach:** The teacher's role is to guide students in performing the calculations for work and power. They will help students understand the formulas ($\text{Work} = \text{Force} \times \text{Distance}$ and $\text{Power} = \text{Work} \div \text{Time}$) and interpret their results to answer the "Think About It" questions.

- **Assessor of Application:** For the "Apprentice" activity, the teacher's primary role is to assess students' individual understanding. They will evaluate whether students can correctly apply the concepts of action-reaction forces from the previous lessons to new scenarios.
- **Facilitator of Reflection:** The teacher will lead a discussion after both activities to connect the concepts. They will use the "Think About It" and "Think and Reflect" sections to help students see the relationship between force, work, power, and the principles of Newton's Third Law.

Expected From Students:

- **Active Experimentation:** For the "Builder" activity, students are expected to actively participate in the experiment, carefully measuring force and time for each box. They must accurately record their data in the provided table.
- **Independent Calculation:** Students are expected to use the provided formulas to calculate work and power for each trial. This demonstrates their ability to apply mathematical concepts to scientific data.
- **Critical Analysis and Connection:** Students should answer the reflection questions in both activities. For "Builder," they need to analyze how weight, force, and time affect work and power. For "Apprentice," they must analyze force diagrams and scenarios to demonstrate their understanding of action-reaction pairs.
- **Individual Assessment:** Students are expected to complete the "Apprentice" worksheet independently, showing their personal mastery of Newton's Third Law and force diagrams.
- **Conceptual Synthesis:** Ultimately, students should be able to connect the two activities—seeing how the forces they measured and calculated in the "Builder" activity are also governed by the action-reaction principles they identified in the "Apprentice" assessment.

Parts of the book included.

3. Now Switch Roles

- Partner B pushes, and Partner A resists.
- Again, notice how it feels to push and resist.

Think about it: What did you feel?

- ☐ I felt a push back when I pushed.
- ☐ My body stayed still.
- ☐ My body moved backward.
- ☐ It was easier when I pushed gently.
- ☐ Other (describe in your own words):

Figure 8: Push-and-Resist Partner Activity to Explore Forces

Think about it: Where is the force going? What pushes back?

Screen Lessons

Let's click on the link to learn more about Newton's Third Law.

<https://www.bekit3UJUR58VvXstGQHEUS0Bgk4tkF>

Assessment

Apprentice

For the following scenarios:

- Match real-life motion scenarios with the correct force diagrams.
- Explain which forces are acting — and reacting — in each case!

Scenario A: A person pushes a walker forward while walking.

Scenario B: A basketball hits the ground and bounces upward.

Figure 9: Different Real-Life Force Scenarios

Scenario C: Two ice skaters push off each other and glide in opposite directions. Scenario D: A person jumps up from a diving board.

Match each scenario with the diagram (A-F) that best shows the forces.

Diagram **Description**

1 → Skater A pushes Skater B; ← Skater B pushes back.

2 → Hands push on walker; ← Walker pushes back on hands.

3 ↓ Ball hits ground; ↑ Ground pushes ball up.

4 ↓ Feet push on board; ↑ Board pushes person up.

Scenario	For Force Diagram No.	Action Force	Reaction Force	Draw force diagram
A				
B				
C				
D				

Think and Reflect: Choose one match and answer:

- Which part of the diagram helped you understand the forces best?
- What surprised you about how the forces worked together?
- Can you think of another real-life example of an action-reaction force pair?



Apprentice

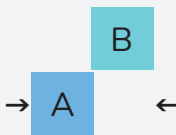
Objective:

Students will demonstrate their understanding of Newton's Third Law by matching real-life scenarios to their corresponding force diagrams and correctly identifying the action-reaction forces. They will also apply critical thinking to reflect on the concepts, articulate their key takeaways, and provide a new example of an action-reaction force pair.

Answers:

- Read each scenario in the list below.
- Review the 4 force diagrams that describe pairs of forces.
- Match each movement to the diagram that best fits it.
- Explain the action and reaction forces involved.
- Draw force Diagrams
- Answer with them reflection questions

Model Answer:

Scenario	Force Diagram No.	Action Force	Reaction Force	Draw force diagram
A	2	→ Hands push on walker	← Walker pushes back on hands	→  ←
B	3	↓ Ball hits ground	↑ Ground pushes ball up	↓  ↑
C	1	→ Skater A pushes Skater B	← Skater B pushes back	→  ←
D	4	↓ Feet push on board	↑ Board pushes person up	↓  ↑

Answers to Reflection Questions:

Sample Reflection (Scenario A: Person pushing a walker — Diagram 3)

1. Which part of the diagram helped you understand the forces best?

The arrows going in opposite directions really helped me. I could clearly see that when the person pushes the walker forward, the walker pushes back with the same force but in the opposite direction. That made Newton's Third Law easier to understand.

2. What surprised you about how the forces worked together?

I was surprised that even though the person is the one pushing, the walker actually pushes

back too! I didn't realize that everything we push also pushes us back — even if we can't always feel it.

3. Can you think of another real-life example of an action-reaction force pair?

Yes — when I jump off a swing. My legs push down on the seat (action), and the swing pushes me up into the air (reaction). That's why I go higher.

Possible Strategies for Implementation

Step 1: Explain the Activity and Set the Task

Strategy: "Structured Matching with Think Time"

How to Implement:

- Display or read aloud each of the 4 scenarios and diagrams. (These are embedded in the student book.)
- Explain that the diagrams show force arrows representing action and reaction.
- Guide them through one example together.
- Provide the matching table for independent or team completion.

Teacher Language: "You're going to think like scientists — matching the motion to what you can't see: the forces. Every arrow in a force diagram tells a story."

Step 2: Matching and Diagram Analysis

Strategy: "Diagram Detective"

How to Implement:

- Students complete the chart by matching each Scenario (A–D) with the correct Diagram (1–4).
- In the table, they write in the action and reaction forces they observe.
- Circulate to ask questions:
 - "How do you know that's the reaction?"
 - "What would happen if there were no reaction?"

Step 3: Drawing force diagrams

Strategy: "Arrow Talk" – Visualizing the Invisible

Implementation Steps:

1. Model with One Scenario (e.g., pushing a walker)

- On the board or projector, draw:
 - A stick figure pushing a walker
 - A red arrow pointing from the person's hands to the walker (label: "Push")
 - A blue arrow pointing from the walker back to the person (label: "Pushes back")

Say aloud: "We draw arrows to show force. The longer the arrow, the stronger the force. Forces always come in pairs — when something pushes, it also gets pushed back."

2. Anchor the Diagram with Two Questions:

- “Who is pushing?” → That’s your action force
- “What pushes back?” → That’s your reaction force

Students apply this thinking to draw arrows in opposite directions from different objects in contact.

3. Student Practice – Scaffolded Drawing

- Give students a blank stick-figure sketch for another scenario (e.g., a ball hitting the ground).
- Ask them to add arrows:
 - One for the ball’s force down
 - One for the ground’s force up
 - Label both arrows: Action / Reaction
- For extra support, give them sentence starters:
 - The ball pushes
 - The pushes back on the ball.

Step 4: Reflect and Share

Strategy: “Choose & Explain”

How to Implement:

- Ask students to pick one of their matched pairs to reflect on.
- Prompt questions:
 - “What did this diagram help you understand?”
 - “Did you change your thinking at any point?”
- Students can present a summary verbally, sketch their own diagram, or share in writing.



Builder

Objective:

Students will be able to investigate the relationship between force, work, and power through hands-on activities. They will demonstrate an understanding of action-reaction forces and how they apply to real-world scenarios, while also learning to measure and calculate these physical concepts. This will enable them to analyze how variables like weight and distance affect the energy needed to move an object.

Model Answer:

1. Use a spring scale or force gauge to measure how much force it takes to push three different boxes light, medium, heavy (in newton's, N) for the same distance (2 meters)
2. Use a timer to record how long it takes (in seconds, s).

3. Use your data to calculate:

- $\text{Work} = \text{Force} \times \text{Distance}$
- $\text{Power} = \text{Work} \div \text{Time}$

Box no.	Force (N)	Distance (m)	Time (s)	Work (J = N × m)	Power (W = J ÷ s)
A (Light)	10 N	2 m	4 s	20 J	5 W
B (Medium)	20 N	2 m	5 s	40 J	8 W
C (Heavy)	30 N	2 m	6 s	60 J	10 W

Think About It:

- **Which box took the most energy to push?**
Box C (Heavy) — because it had the highest force and took the most work (60 J).
- **Which one needed the least energy?**
Box A (Light) — it needed only 10 N of force and resulted in the least work (20 J).
- **How do weight and distance affect the work and power?**
Heavier boxes (more weight) need more force to move, which increases the work and power. The more distance you move something, the more work it takes too.
- **How might this help you improve your smart walker later?**
I'll try to reduce the force needed to push the walker (like using better wheels or lighter materials) so it's easier and more efficient for the user.

Materials needed:

- 3 boxes (Light, Medium, Heavy – pre-filled or labeled)
- Spring scale or force gauge
- Measuring tape or meter stick
- Timer or stopwatch
- Calculator

Step by step instructions:

1. Label Your Boxes as:

■ A = Light, ■ B = Medium, ■ C = Heavy

2. Attach the Spring Scale:

- Hook the spring scale to Box A. Pull the box at a steady speed.
- Record the force reading in Newtons (N) when it moves smoothly.

3. Measure the Distance:

- Use a tape measure to record how far the box moves.

Tip: Keep distance consistent (e.g., 2 meters).

4. Time It!

- Use a stopwatch to record how many seconds it takes to move the box over the measured distance.

5. Record Your Data:

- Write down the force, distance, and time in the table provided in your book.

6. Repeat for Boxes B & C:

- Follow steps 2 to 5 for the medium and heavy boxes.

7. Calculate Work & Power:

- Use these formulas:
 - $\text{Work} = \text{Force} \times \text{Distance}$
 - $\text{Power} = \text{Work} \div \text{Time}$
- Use a calculator and write your answers in the table!

8. Reflect:

- Answer the “Think About It” questions to explain your understanding.

Possible Strategies for Implementation:

Strategy 1: Modeling First

Implementation:

- Start by modeling one trial in front of the class.
- Show how to use the spring scale, measure distance (2 meters), time the push, and fill in one row in the table.
- Talk aloud your thought process:
“I’m pulling this box — it says 10 N on the scale. Let’s see how far it moves... and how much time it takes... Now let’s do the math together.”

Strategy 2: Station-Based Collaborative Work

Implementation:

- Divide students into small mixed-ability groups.
- Assign roles: measurer, timer, recorder, calculator.
- Rotate through 3 box weights (light, medium, heavy) placed at clearly marked stations.

Strategy 3: Scaffolded Data Recording

Implementation:

- Use color-coded box labels (e.g., red = heavy, green = light).
- Provide pre-filled templates for force/distance columns for students who may need support.
- Allow calculator use or peer assistance for power/work formulas.

Strategy 4: Reflect and Relate

Implementation:

- After completing the table, pose the thinking questions.
- Let students choose how to respond:
 - Write answers
 - Record a voice note
 - Sketch a comparison (e.g., energy arrows for each box)

Sample Teacher Prompt: “Now that you’ve felt the difference in effort between the boxes, let’s connect this to designing something real — like a walker. What would be the problem with needing too much force?”



Mastermind

Objective:

Students will apply their understanding of Newton's Third Law and its real-world applications by observing a daily object, analyzing its design for comfort and accessibility, and proposing a redesign for a user with specific needs.

Activity instructions:(This is a home work activity)

- 1. Choose Your Object:** Walk around your home and select one everyday object. This could be a kitchen tool (like a can opener or a whisk), a piece of furniture (like a chair or a lamp), or another common item (like a broom or a pair of scissors).
- 2. Observe and Reflect:** Carefully observe how this object is used by different people in your household.
 - Comfort: Is the object comfortable to use for everyone? Does the grip feel good? Is the size or weight appropriate?
 - Fit and Function: Can people of different sizes, ages, or abilities use it easily? For example, can a young child or an elderly person use it without difficulty?
 - Identify a Challenge: Think about the ways the object could be improved, particularly for someone who might have trouble using it.
- 3. Propose a Redesign:** Based on your observations, come up with a creative idea for how you would redesign this object. Your redesign should specifically aim to make it more accessible, comfortable, or functional for an older adult or someone with a physical disability.

4. Document Your Work: On a separate sheet of paper, you must document your process.

- Part A: The Original Object: Write down the name of the object you chose and a few sentences describing how it is typically used and what its current design challenges are.
- Part B: The Redesign Idea: Describe your redesigned object. Be specific! Explain the new features, materials, or shape. For example, would you change the handle? The weight? The color?
- Part C: The Rationale: Explain why your redesign would be better for an older adult or a person with a disability. Connect your ideas back to what you learned about forces, grip, and comfort. You can also include a simple sketch of your redesign to help explain your idea.

Final Task: Be prepared to present your redesign idea to the class and explain the reasoning behind your choices.



Now I Can

Objective:

Students will be able to demonstrate an understanding of Newton's Third Law by explaining its application in movement and in the design of a Smart Walker. They will also be able to reflect on their learning journey, articulate what they can now do, and identify areas for future growth and application.

Step 1: Student Self-Reflection

- Students will individually read through the "Now I Can" statements. They will think about which skills they have gained from the previous activities, such as the "Let's Think!", "Screen lessons," and "DIY Lab," and honestly assess their own confidence level for each one. This is a personal check-in, not a graded assignment.

Step 2: Small Group Discussion

- Students will form small groups to discuss their individual reflections. They will share which "Now I Can" statements they found easiest or most challenging and explain their reasons. This step allows for peer learning and provides an opportunity for students to help each other understand areas where they might be struggling.

Step 3: Celebrate Growth

- The entire class will come back together for a final discussion. The teacher will ask for volunteers to share something new they learned during the unit. The teacher will then connect this new knowledge to future lessons, showing how their current progress is a foundation for what's to come, and will conclude by celebrating the effort and progress of every student.

Relation to Standards

1. Science Standards

- **Forces and Interactions (Physical Science):** The core of the lesson directly addresses Newton's Third Law of Motion, which states that for every action, there is an equal and opposite reaction. The activities "DIY Lab" and "Screen lessons" specifically focus on helping students understand and identify these action-reaction pairs in real-world scenarios.
- **Motion and Stability:** The lesson also touches on the concepts of force and motion. Students explore how forces affect an object's motion and stability (e.g., how a walker helps maintain balance) and how to measure and calculate these forces.

2. Engineering & Technology Standards

- **Engineering Design Process:** The "Mastermind" activity is a perfect example of the engineering design process. Students are asked to:
 - **Identify a Problem:** Analyze a daily object and identify a design flaw for a specific user (e.g., an older adult).
 - **Propose a Solution:** Brainstorm and propose a redesign.
 - **Synthesize Knowledge:** Apply scientific concepts (like forces and stability) to justify their design choices.
- **Analyzing Design:** Students are required to critically analyze the function and form of a real-world object and think about its user-friendliness, which is a core skill in design and technology.

3. Math Standards

- **Measurement and Data Analysis:** The "Builder" activity, which involves measuring force and time, and then calculating work and power, directly addresses standards related to data collection, measurement, and mathematical problem-solving.
- **Formula Application:** Students learn to apply and understand simple formulas ($\text{Work} = \text{Force} \times \text{Distance}$) to solve real-world problems.
- **Qualitative and Quantitative Reasoning:** The lesson moves from qualitative observations ("What did it feel like?") in the "DIY Lab" to quantitative measurements in the "Builder" activity, reinforcing the connection between observation and data.